Rank Transformations as a Bridge Between Parametric and Nonparametric Statistics

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Many of the more useful and powerful nonparametric procedures may be presented in a unified manner by treating them as rank transformation procedures. Rank transformation procedures are ones in which the usual parametric procedure is applied to the ranks of the data instead of to the data themselves. This technique should be viewed as a useful tool for developing nonparametric procedures to solve new problems.

KEY WORDS: Nonparametric; Rank transformation; Regression; Experimental designs; Discriminant analysis; Multiple comparisons

1. INTRODUCTION

A problem that applied statisticians have been confronted with virtually since the inception of parametric statistics is that of fitting real world problems into the framework of normal statistical theory when many of the data they deal with are clearly nonnormal. From such problems have emerged two distinct approaches or schools of thought: (a) transform the data to a form more closely resembling a normal distribution framework or (b) use a distribution free procedure. The first method may include the log transformation, square root transformation, arcsin transformation, and so forth, and may even be broad enough to include robust procedures that tend to give small weights to outliers, that is, to observations that may contribute greatly to the nonnormal form of the data. The second method includes a large body of methods based on the ranks of the data.

There is a way of combining these two methods by presenting many nonparametric methods as parametric methods applied to transformed data. Simply replace the data with their ranks, then apply the usual parametric t test, F test, and so forth, to the ranks. We call this the rank transformation (RT) approach. This approach results in a class of nonparametric methods that includes the Wilcoxon-Mann-Whitney test, the Kruskal-Wallis test, the Wilcoxon signed ranks test, the Friedman test, Spearman's rho, and others. The rank transformation approach also furnishes useful methods in multiple regression, discriminant analysis, cluster analysis, analysis of experimental designs, and multiple comparisons.

Of course, there are several ways in which ranks can be assigned to observations. We suggest the following types.

RT-1. The entire set of observations is ranked from smallest to largest, with the smallest observation having rank 1, the second smallest rank 2, and so on. Average ranks are assigned in case of ties.

RT-2. The observations are partitioned into subsets and each subset is ranked within itself independently of the other subsets.

RT-3. This rank transformation is RT-1 applied after some appropriate reexpression of the data.

RT-4. The RT-2 type is applied to some appropriate reexpression of the data.

The rank transformation approach provides a useful pedagogical technique for introducing these nonparametric methods as an integral part of an introductory course in statistics, instead of isolating the methods in a separate unit that may appear to the student to be disconnected from the general flow of the course. Also, it allows the practitioner to make full use of existing statistical packages that may not have suitable nonparametric programs by simply entering the ranks of the data into the programs for the parametric analysis. And finally, this approach may be viewed as a useful tool for developing new nonparametric methods in situations where satisfactory parametric procedures exist.

2. TWO INDEPENDENT SAMPLES

Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ represent two independent random samples. To test the hypothesis that $E(X) = E(Y)$ the parametric procedure employs the two-sample t statistic

$$t = \frac{\bar{X} - \bar{Y}}{\left[\left(\sum (X_i - \bar{X})^2 + \sum (Y_i - \bar{Y})^2\right) \dfrac{N}{nm(N-2)}\right]^{1/2}} \tag{2.1}$$

where $N = n + m$, and compares t with quantiles from the t distribution with $N - 2$ degrees of freedom (df). The nonparametric Wilcoxon-Mann-Whitney two-sample test requires replacing the data by the ranks R_i from 1 to N, and uses the statistic, in its standardized form with the adjustment for ties incorporated,

$$T = \frac{S - n(N+1)/2}{\left[\dfrac{nm}{N(N-1)} \sum_{i=1}^{N} R_i^2 - \dfrac{nm(N+1)^2}{4(N-1)}\right]^{1/2}} \tag{2.2}$$

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where $S = \sum_{i=1}^{n} R_i$ is the sum of the ranks of the X's. The statistic T is compared with the standard normal distribution or, if there are no ties and the sample sizes are less than 20, exact tables may be used for S (Conover 1980).

A rank transformation procedure is based on computing t on the ranks R_i to get the statistic

$$t_R = \frac{1}{n} S - \frac{1}{m}\left(\frac{N(N+1)}{2} - S\right) \div \left[\left(\sum_{i=1}^{N} R_i^2 - \frac{1}{n} S^2 - \frac{1}{m}\left(\frac{N(N+1)}{2} - S\right)^2\right)\frac{N}{nm(N-2)}\right]^{1/2} \quad (2.3)$$

and using the t tables as with (2.1). This is an example of an RT-1 type procedure. A little algebra reveals an important relationship between t_R and T:

$$t_R = \frac{T}{\left[\frac{N-1}{N-2} - \frac{1}{N-2} T^2\right]^{1/2}}, \quad (2.4)$$

which shows that t_R is a monotonically increasing function of T. Since T contains the correction for ties, so does t_R because it is a function of T.

Let us consider the implication of (2.4). When T is in its upper α level tail region, then t_R is in its upper α level tail region also. The same can be said for the lower α level tail regions of each. For example, when $n = 14$ and $m = 18$ the upper five percent value for S (Conover 1980, Table A.7) is 274 if there are no ties. Substitution of S into (2.2) and (2.3) reveals the exact upper five percent values for T and t_R to be 1.633 and 1.681, respectively. When T is compared with the upper five percent quantile 1.645 from a standard normal distribution, a slightly conservative test results. The t distribution with 30 df gives an upper five percent critical value of 1.697, also resulting in a slightly conservative test for t_R. Because t_R is a monotonic function of T, the two tests are equivalent when the exact critical values are used. The normal distribution and the t distribution provide two different approximations, which have been compared by Iman (1976). The Wilcoxon-Mann-Whitney test, with all of its good properties, may be performed using t_R as a test statistic instead of T if desired. In fact t_R may be preferred, as computing routines are generally readily available for the t statistic; and also it may be simpler to teach this procedure to someone who understands the t test and transformations, but who does not have a working knowledge of nonparametric statistics.

3. k INDEPENDENT SAMPLES

Consider k independent random samples, $(X_{11}, \ldots, X_{1n_1}), \ldots, (X_{k1}, \ldots, X_{kn_k})$ and let

$$SSA = \sum_{i=1}^{k} n_i(\bar{X}_i - \bar{X}_{..})^2 \quad (3.1)$$

and

$$SSE = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (\bar{X}_{ij} - \bar{X}_i)^2 \quad (3.2)$$

represent the analysis of variance sums of squares. The usual parametric test of the hypothesis of equal means compares the statistic

$$F = (SSA/(k-1))/(SSE/(N-k)) \quad (3.3)$$

with the F distribution, $k - 1$ and $N - k$ df.

For the Kruskal-Wallis test the data are replaced by their ranks $R(X_{ij})$ from 1 to $N = \sum n_i$ and the statistic, incorporating the correction for ties, is given as

$$H = \frac{\sum_{i=1}^{k} R_i^2/n_i - N(N+1)^2/4}{(\sum_i \sum_j R^2(X_{ij}) - N(N+1)^2/4)/(N-1)} \quad (3.4)$$

where

$$R_i = \sum_{j=1}^{n_i} R(X_{ij}).$$

The statistic H is compared with the chi-squared distribution, $k - 1$ df. The most extensive tables of the exact distribution of H, applicable only if there are no ties, are by Iman, Quade, and Alexander (1975).

A rank transformation procedure is based on computing F on the ranks to get

$$F_R = \frac{\left[\sum_{i=1}^{k} \frac{R_i^2}{n_i} - \frac{N(N+1)^2}{4}\right] \Big/ (k-1)}{\sum_{i=1}^{k} \sum_{j=1}^{n_i} \left(R(X_{ij}) - \frac{R_i}{n_i}\right)^2 \Big/ (N-k)} \quad (3.5)$$

as a test statistic. This is another example of an RT-1 type procedure. Elementary algebra reveals F_R is a monotonic function of H,

$$F_R = (H/(k-1))/((N-1-H)/(N-k)) \quad (3.6)$$

so the two tests are equivalent. The upper α level critical value from the chi-squared distribution, when substituted for H in (3.6), results in a slightly different critical value for F_R than that obtained from the appropriate tables of the F distribution. Both methods, however, merely provide approximations to the true critical value. Iman and Davenport (1976) compare these and other approximations and show that the F approximation should be preferred to the chi-squared in most cases.

4. THE ONE-SAMPLE OR MATCHED-PAIRS PROBLEM

Let $D_1, \ldots, D_n$ represent independent random variables with a common mean where, in the case of matched pairs (X_i, Y_i), D_i equals $X_i - Y_i$. To test the hypothesis $E(D) = 0$ the one-sample t statistic

$$t = \frac{\sum D_i}{\left[\frac{n}{n-1} \sum D_i^2 - \frac{1}{n-1} (\sum D_i)^2\right]^{1/2}} \quad (4.1)$$

is compared with the t distribution, $n - 1$ df, in the parametric test valid for normally distributed D's.

For the Wilcoxon signed ranks test the D_i's are replaced by the signed ranks R_i, where

$$R_i = (\text{sign } D_i) \times (\text{rank of } |D_i| \text{ among } |D_1|, \ldots, |D_n|). \quad (4.2)$$

The hypothesis is rejected when the test statistic

$$T = (\sum R_i)/(\sqrt{\sum R_i^2}) \quad (4.3)$$

is too large or too small, as measured by the normal approximation. If n is small and there are no ties, exact tables may be used (cf. Conover 1980, Table A.13). The statistic T is equivalent to the form for this test, which appears in most textbooks and is based on the sum of the positive ranks only. This form is simpler to use in the presence of ties, since the correction for ties is incorporated into (4.3).

Alternatively, the one-sample t statistic may be computed on the signed ranks, resulting in

$$t_R = \frac{\sum R_i}{\left[\frac{n}{n-1}\sum R_i^2 - \frac{1}{n-1}(\sum R_i)^2\right]^{1/2}} \quad (4.4)$$

which is compared with the t distribution, $n - 1$ df, as an approximation. Since D_i represents a reexpression of the data (X_i, Y_i) and may be considered to be the product of (sign D_i) and $|X_i - Y_i|$, this is an example of an RT-3 type procedure.

Note that t_R is also expressible as

$$t_R = \frac{T}{\left[\frac{n}{n-1} - \frac{1}{n-1}T^2\right]^{1/2}}, \quad (4.5)$$

which is a monotonic function of T. Thus the test that uses t_R is equivalent to the test that uses T. A comparison of the normal approximation, the student's t approximation, and the exact distribution is given by Iman (1974a).

Suppose t in (4.1) is applied directly to RT-1 type ranks; that is, the X's and Y's are replaced by their corresponding ranks 1 to $2n$ and D_i is the difference in those ranks. This application of the rank transformation approach does not yield the Wilcoxon signed ranks test, but rather introduces a new procedure. This new procedure is conditionally distribution free given the ranks in the blocks and asymptotically distribution free by virtue of the central limit theorem. Properties of this test are reported by Iman and Conover (1980a).

5. THE RANDOMIZED COMPLETE BLOCK DESIGN

In the randomized complete block design with one observation per cell, let X_{ij} be the random variable for block i, treatment j, $i \le b$, and $j \le k$. If $\bar{X}_{i\cdot}$, $\bar{X}_{\cdot j}$, and $\bar{X}_{\cdot\cdot}$ represent the block, treatment, and grand means respectively, then

$$SST = b \sum_{j=1}^{k} (\bar{X}_{\cdot j} - \bar{X}_{\cdot\cdot})^2 \quad (5.1)$$

and

$$SSE = \sum_{i=1}^{b} \sum_{j=1}^{k} (X_{ij} - \bar{X}_{\cdot j} - \bar{X}_{i\cdot} + \bar{X}_{\cdot\cdot})^2 \quad (5.2)$$

are the analysis of variance treatment and error sums squares. The parametric test of equal treatment effects compares the statistic

$$F = (SST/(k-1))/(SSE/(b-1)(k-1)) \quad (5.3)$$

with the F distribution, $k - 1$ and $(b - 1)(k - 1)$ df.

The usual nonparametric test involves ranking the observations from 1 to k within each block, making no interblock comparisons. The Friedman test uses the statistic, corrected for ties,

$$T = \frac{(k-1)\sum_{j=1}^{k}[R_j - b(k+1)/2]^2}{\sum_i \sum_j R^2(X_{ij}) - bk(k+1)^2/4} \quad (5.4)$$

where R_j is the sum of the ranks $R(X_{ij})$ for treatment j. The chi-squared distribution with $k - 1$ df is used as an approximation to the distribution of T.

Another way of considering the Friedman test is to compute the F statistic in (5.3) on the intrablock ranks that are used in the Friedman test. This is a type RT-2 procedure. The result is a statistic F_R that is a monotonic function of the Friedman statistic

$$F_R = (T/(k-1))/((b(k-1) - T)/(b-1)(k-1)). \quad (5.5)$$

Comparison of F_R with the F distribution provides a more accurate approximation (Iman and Davenport 1980) than the chi-squared approximation used with (5.4).

Suppose F is applied directly to RT-1 type ranks where all of the observations are ranked together, from 1 to bk in this case. This type of ranking takes advantage of both between and within block information. The result is a test which is conditionally distribution free, given the partitioning of ranks into blocks. This procedure, with the F distribution as an approximation, compares favorably with the RT-2 type Friedman test (Iman and Conover 1980a), and even Fisher's randomization test (Conover and Iman 1980a) in terms of robustness and power.

It is easy to extend the RT-1 type procedure to other experimental designs. This approach is robust and powerful in the two-way layout with interaction (Iman 1974b), in a test for interaction when replication effects are present (Conover and Iman 1976), and in a test for main effects in the presence of replication and interaction effects (Iman and Conover 1976).

The advantage of ranking all of the observations together is that any analysis of variance procedure

may be applied to the ranks, with the resulting tests for main effects, interactions, or whatever, following immediately. Other rank tests that involve a separate ranking for each test of hypothesis become difficult or impossible to apply. The same may be said for aligned ranks tests, in which the appropriate means are first subtracted from each observation before ranking. This resembles an RT-3 type procedure. In addition, for aligned ranks tests some power may be lost in the process (Conover and Iman 1976).

6. CORRELATION

One of the earliest applications of a rank transformation involves computing Pearson's product moment correlation coefficient

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{[\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2]^{1/2}} \tag{6.1}$$

on ranks to obtain Spearman's rho

$$\rho = \frac{\sum \left(R(X_i) - \frac{n+1}{2}\right)\left(R(Y_i) - \frac{n+1}{2}\right)}{\left[\sum \left(R(X_i) - \frac{n+1}{2}\right)^2 \sum \left(R(Y_i) - \frac{n+1}{2}\right)^2\right]^{1/2}} \tag{6.2}$$

for paired data $(X_1, Y_1), \ldots, (X_n, Y_n)$. Since the observations within the subset $\{X_i\}_{i=1}^n$ are ranked within themselves, and the same is true for the subset $\{Y_i\}_{i=1}^n$, this is an example of an RT-2 type procedure. Just as r is a measure of linearity of the relationship between X and Y, so is ρ a measure of the linearity between the ranks of X and the ranks of Y, which translates as a measure of monotonicity in the relationship between X and Y.

The direct extension to multiple correlation is obvious. The observations on each component X_{ij} of $\mathbf{X}_j = (X_{1j}, \ldots, X_{kj}), j = 1, \ldots, n$ are ranked separately from 1 to n. Multiple correlations, partial correlations, and the like may be computed on the ranks just as they would be computed on the data.

To test for independence between X and Y the statistic

$$t = r\sqrt{n-2}/\sqrt{1-r^2} \tag{6.3}$$

is compared with the student's t distribution with $n - 2$ df, in a parametric test valid with bivariate normal distributions. The nonparametric test statistic takes the form

$$Z = \rho\sqrt{n-2}, \tag{6.4}$$

which is compared with the standard normal distribution.

The computation of t on the rank transformed observations results in the statistic

$$t_R = \rho\sqrt{n-2}/\sqrt{1-\rho^2}, \tag{6.5}$$

which is compared with the same distribution used with (6.3). This approximation was suggested by Pitman (1937). A comparison of the normal approximation on Z with the student's t approximation on t_R by Iman and Conover (1978) shows the latter approximation to be slightly better.

7. REGRESSION

The adaptation of correlation procedures to the ranks of the data immediately suggests adapting regression methods to RT-2 type data. Least squares, forward or backward stepwise regression, or any other regression method may be applied to the ranks of the observations (Iman and Conover 1979). The result is that the rank of the dependent variable is predicted using the ranks of the independent variables. The predicted rank $\hat{R}(Y_i)$ may be transformed back into a predicted value $\hat{Y}$ of Y, by linear interpolation between the two values of Y that have ranks bracketing $\hat{R}(Y_i)$.

Another way of using a rank transformation to develop nonparametric methods in regression is to assume the regression equation is linear and use ordinary least squares to obtain an estimated regression equation $Y = a + bx$. The parametric method of testing the hypothesis $\beta = \beta_0$ for slope uses the statistic

$$t = \frac{(b - \beta_0)[(n-2) \sum (X_i - \bar{X})^2]^{1/2}}{[\sum (Y_i - \bar{Y} - b(X_i - \bar{X}))^2]^{1/2}}, \tag{7.1}$$

which has a t distribution under certain assumptions including normality. It is easier to see how a rank transformation leads to a nonparametric test if t is rearranged as follows:

$$t = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y} - \beta_0(X_i - \bar{X}))\sqrt{n-2}}{[\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y} - b(X_i - \bar{X}))^2]^{1/2}}. \tag{7.2}$$

Note that, except for a multiplicative constant, the numerator is the sample covariance of X and the hypothesized residuals, and the denominator has the sample variances of X and the least squares residuals. Replace the covariance with the covariance between the ranks of X and the ranks of the hypothesized residuals, which is an RT-4 type transformation. Use the variances of those ranks in the denominator, and one has the nonparametric test for slope proposed by Hogg and Randles (1975).

A natural extension of the rank transformation approach to the general linear model consists of ranking each quantitative variable separately in the general linear model and applying usual parametric procedures. This is an RT-2 type application. The result is new tests for equal slopes, analysis of covariance (Conover and Iman 1980b), and any experimental designs covered by the general linear model. These tests are in general not distribution free, except perhaps in some asymptotic sense, but they may be more robust and powerful than their competitors in nonnormal situations. Each procedure, however, needs to

be evaluated on its own merits. Another RT-2 type procedure for analysis of covariance is given by Quade (1967).

8. DISCRIMINANT ANALYSIS

The usual linear discriminant function (LDF) and quadratic discriminant function (QDF) have been applied to the RT-2 type data with good results. In an extensive simulation study (Conover and Iman 1980c) these rank transformation methods showed an ability to discriminate between populations that equaled the LDF and QDF procedures with normal populations, and surpassed them with nonnormal populations. Unlike the nonparametric competitors, these methods can be used with small samples. Also no new computer programs are required, since routines with LDF and QDF are readily available. All that is required is an RT-2 type transformation of the data before entering the data into the computer.

9. MULTIPLE COMPARISONS

Any of the popular multiple comparisons techniques, including Scheffé's, Tukey's, Duncan's, and Fisher's methods, as well as others, may be applied to the RT-1 type data with good results. The power with normal populations is about the same whether the analysis is done on the data or on the ranks. With nonnormal populations the multiple comparisons procedures are more robust and have more power when rank transformed data are used (Iman and Conover 1980b).

10. BIOASSAY

Williams (1971, 1972) developed a procedure designed to compare the toxicity of various dose levels of a drug against the zero dose level. This same procedure is used on the RT-1 type data by Shirley (1977) in a nonparametric test.

11. CLUSTER ANALYSIS

Scott and Knott (1974) developed a method of partitioning means in the analysis of variance into two clusters. The same procedure is applied to the RT-1 type data to obtain a distribution free procedure by Worsley (1977).

12. DISCUSSION

Nonparametric methods should be among the working tools of any statistician. The rank transformation approach provides a vehicle for presenting both the parametric and nonparametric methods in a unified manner. This should enable novice statisticians to understand the differences and similarities of the two types of analysis. Also the rank transformation approach leads to easier computational methods, since it is often more convenient to enter ranks into a program for parametric analysis than it is to find or write a program for a nonparametric analysis. Most existing programs for nonparametric methods do not incorporate corrections for ties, while rank transformation procedures all automatically make the required corrections for ties. The user merely uses average ranks whenever ties occur.

Other scores may be used instead of ranks, if desired, to obtain nonparametric tests that are equivalent to tests such as the van der Waerden test, Capon test, median test, McNemar test, and others. Our experience indicates that the ranks themselves provide scores that are difficult to improve upon for general all-around use.

Some limitations of the rank transformation approach should be noted here also. The rank transformation procedures lead to distribution free tests in some cases, while in other cases the resulting tests may be only conditionally distribution free, asymptotically distribution free, or neither. An example of the latter case arises when the ratio of two sample variances is computed on RT-1 type data and compared with tables of the F distribution as a test for equal variances in two independent samples. Simulation results indicate a severe lack of robustness for this procedure, even when the population means are equal. This is probably related to the fact that the parametric F test for this problem is notoriously sensitive to normality, and the central limit theorem does not apply in this situation. A referee pointed out that a similar situation exists when Welch's t test is used on ranks of the data in the hopes of solving the nonparametric Behrens-Fisher problem.

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Comment

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To gain greater generality, statisticians have applied standard normal theory procedures to the ranks of observations for a long time. Conover and Iman have now systematized this approach in a series of papers. The present survey provides a readable, nontechnical account of this work.

The authors call attention to three potential uses of the method of rank transformations: (a) as a pedagogical technique for incorporating nonparametrics in introductory statistics courses; (b) as a device for adapting normal-theory computer packages to the computation of nonparametric statistics; and (c) as a tool for developing new nonparametric procedures in situations where parametric procedures already exist. My concern is primarily with the first of these three suggested uses.

During the last 30 years or so the role played by nonparametrics has undergone considerable change. In the 1950's, and even the 1960's, it was quite common to characterize nonparametric methods as "rough and ready." Few practicing statisticians were inclined to accept them into their regular arsenal of statistical tools. This attitude is clearly changing. Now practically every introductory statistics text contains a chapter concerning nonparametric, or distribution-free, or rank methods. As Conover and Iman point out, "Nonparametric methods should be among the working tools of any statistician."

The question is no longer whether but how nonparametrics should be incorporated in an introductory statistics course. Conover and Iman advocate that nonparametric methods be taught within the normal-theory framework. I should like to give three reasons for opposing this point of view.

1. It unnecessarily restricts the perceived applicability of the nonparametric procedure.

2. It further emphasizes the widely held misconception that the nonparametric approach is essentially restricted to hypothesis testing.

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3. And what is most serious to my way of thinking, it completely hides the greater conceptual and theoretical simplicity of many nonparametric procedures.

The two-sample problem discussed by Conover and Iman as their introductory example illustrates all three points.

To derive the Wilcoxon-Mann-Whitney (WMW) test, Conover and Iman start from the two-sample t statistic for testing the hypothesis $E(X) = E(Y)$. In the appropriate normal-theory framework, the t statistic is appropriate for problems involving the shift parameter $\Delta = E(Y) - E(X)$. As a result, the WMW test will be perceived as a test for shift in location. Though quite common, this is much too restricted a view of the WMW test, as my subsequent discussion clearly shows.

With respect to my second objection, even if we should be interested in the shift model, the rank transformation approach only tells us how to test the hypothesis $\Delta = 0$. It gives no indication of how we can estimate Δ.

In connection with my third objection, I am not concerned about the serious statistician who has a good understanding of the nature of a t statistic. I am concerned about the several hundred thousand students who each year are exposed to a single introductory statistics course. For most, the t statistic is just another formula. Substituting ranks in this formula will have even less meaning than substituting actual observations, and very likely will raise doubts in their minds about the need for any assumptions.

Is there an elementary approach to the Wilcoxon test that covers estimation as well? Something like the following has worked fairly well for me.

Suppose we want to investigate a claim that type B batteries have 50 percent longer service life than type A batteries. If Y stands for the service life of a type B battery, and X for that of a type A battery, we can set up the model

$$Y = \tau X, \tag{1}$$

where τ is an unknown scale parameter. Our first question is: How can we estimate τ?

Given just one measurement X_1 for type A and one measurement Y_1 for type B, the obvious point estimate of τ is $Y_1/X_1 = E_{11}$, say. If there is a second Y measurement Y_2, we can form the additional estimate $E_{21} = Y_2/X_1$ and make the following claim,

$$E_{(1)} < \tau < E^{(1)}, \tag{2}$$

where $E_{(1)}$ is the smaller and $E^{(1)}$ is the larger of the two estimates E_{11} and E_{21}. Simple enumeration shows that interval (2) has confidence coefficient 2/6.

For samples $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$, there are *mn elementary estimates* $E_{ji} = Y_j/X_i, j = 1, \ldots, m$, $i = 1, \ldots, n$, and in generalization of (2), we can consider confidence intervals

$$E_{(d)} < \tau < E^{(d)}, \tag{3}$$

where d is a positive integer, $E_{(d)}$ is the dth smallest, and $E^{(d)}$ the dth largest of these mn elementary estimates. For increasing d, the intervals (3) narrow down on the median that we use as our point estimate,

$$\hat{\tau} = \text{median } E_{ji}.$$

Now we are ready to tackle the original problem: do type B batteries provide 50 percent longer service life than type A batteries? Or, in terms of model (1), is 1.5 an acceptable value for τ? If the value 1.5 falls in the confidence interval (3), we accept the claim; if it does not, we reject the claim.

Experimenters are most often interested in the hypothetical value $\tau = 1$, which implies no differences between X and Y measurements. The hypothesis $\tau = 1$ can be tested without computing the mn elementary estimates E_{ji}. The hypothetical value $\tau = 1$ is not in the confidence interval (3) if either

$$E_{(d)} > 1 \quad \text{or} \quad E^{(d)} < 1.$$

In the first case, $U = \#(E_{ji} < 1) = \#(Y_j < X_i) < d$. In the second case, $U' = \#(E_{ji} > 1) = \#(Y_j > X_i) < d$. The two-sample hypothesis is rejected, if the smaller of U and U' is smaller than d. This is a very useful form of the Mann-Whitney test.

Not only have we derived the Mann-Whitney test, and therefore the Wilcoxon test, by completely elementary arguments involving no more complicated probability considerations than equally likely cases, we have shown at the same time that the test is appropriate for the scale model (1) and we even know how to estimate the scale parameter τ using a point estimate or a confidence interval.

It should be obvious that a completely parallel development that starts from the shift model

$$Y = X + \Delta$$

and uses elementary estimates $E_{ji} = Y_j - X_i$ leads to the same Mann-Whitney test. Either development shows that in computing U, we are really trying to estimate the probability that a Y measurement is smaller than an X measurement. This fundamental aspect of the WMW test is completely hidden by the rank transformation approach.

Similarly elementary developments are possible for the sign and Wilcoxon signed rank tests as well as several statistical procedures based on Kendall S.

Comment

MICHAEL A. FLIGNER*

In their article, Conover and Iman describe three areas in which rank transformations can be useful, namely, (a) as a pedagogical technique for introducing nonparametric methods in an introductory statistics course; (b) as a method for using existing statistical packages to compute nonparametric statistics; and (c) as a useful tool for developing new nonparametric methods in situations where satisfactory parametric procedures exist.

My views on (a) are similar to those of Noether, and I feel there can be no disagreement with (b), *provided* one is using a rank transformation procedure known to have good properties. My main concern is with area (c). There are many problems in areas such as linear models, multivariate methods, and time series analysis, to name a few, for which satisfactory parametric procedures exist but for which there is not general agreement on their appropriate nonparametric analogs. Rank transformation methods can provide quick solutions to many of these problems, but the resulting solutions may not be the best, or may not even be appropriate nonparametric methods. In fact, sometimes the problem being solved by the rank transformation method is quite different from that being solved by the original parametric method. It must also be emphasized that rank transformations do not provide a well-defined methodology for developing new nonparametric procedures, as the technique leaves open the questions of what to rank, how to rank, as well as any properties, desirable or otherwise, of the resulting procedures.

I would like to illustrate my points by developing a rank transformation analog to a recently proposed parametric multiple comparison procedure. The resulting solution is to a new problem of questionable interest and is also mathematically incorrect. I feel this example is useful in describing some of the difficulties that can arise when using rank transformations to develop new procedures. In fact, a procedure similar to the one here, which has all its drawbacks, has recently appeared (see Levy 1980).

Briefly, we have $k - 1$ independent random samples $X_{i1}, \ldots, X_{in}$ from treatment populations with cdf's $F(x - \theta_i)$, $i = 1, \ldots, k - 1$, and a sample $X_{k1}, \ldots, X_{km}$ from a control population with cdf $F(x - \theta_k)$. Under the assumption that F is normal, Dunnett (1955, 1964) obtains simultaneous confidence intervals on $\theta_i - \theta_k$, $i = 1, \ldots, k - 1$, while Shaffer (1977) extends this procedure to include simultaneous confidence intervals for all linear contrasts, $\sum_{i=1}^{k} c_i\theta_i$, where $\sum_{i=1}^{k} c_i = 0$. Shaffer's intervals are most sensitive to control versus treatment comparisons, which are of primary interest, while still allowing one to study other types of contrasts.

For a rank transformation analog, first replace X_{ij} by its rank R_{ij} in the combined samples. By using ranks rather than the original observations, one is forced to consider contrasts of the form $\sum_{i=1}^{k} c_i E\bar{R}_{i\cdot}$ as the analog of Shaffer's contrasts $\sum_{i=1}^{k} c_i\theta_i = \sum_{i=1}^{k} c_i E\bar{X}_{i\cdot}$. These new contrasts measure something quite different than do those in the original problem. This is because $E\bar{R}_{i\cdot}$ is a complicated function of the θ's, F, and even m and n, thereby creating what I feel to be a serious difficulty in interpreting the exact meaning of the new contrasts.

A common concern about simultaneous inference procedures using joint rankings (Lehmann 1975, p. 245) is that the *decision* regarding a subset of the populations is dependent on the observations from the other populations. This procedure carries this one step further by also allowing the *parameter* (i.e., $\sum c_i E\bar{R}_{i\cdot}$) used in comparing a subset of the populations to depend on the other populations. Suppose F is normal with mean 0 and variance ½, $\theta_1 = -\theta_2$, and $\theta_k = 0$. When using Shaffer's contrast, the average effect of treatments 1 and 2 will be the same as the control. However, the rank contrast comparing the average effect of treatment 1 and 2 with the control, namely $(ER_{1\cdot} + ER_{2\cdot})/2 - ER_{k\cdot}$, need not be zero, but it will depend on the values of $\theta_3, \ldots, \theta_{k-1}$, F and even m and n. For example, letting $k = 4$, $m = n$, $\theta_1 = \theta_3 = \theta$, $\theta_2 = -\theta$, and $\theta_4 = 0$, one can easily show that

$$(E\bar{R}_{1\cdot} + E\bar{R}_{2\cdot})/2 - E\bar{R}_{4\cdot} = n[(\Phi(-2\theta) + \Phi(0))/2 - \Phi(-\theta)], \quad (1)$$

where Φ is the standard normal distribution function. Expression (1) is not zero unless $\theta = 0$ and also depends on n in an undesirable way. If F is not normal but is symmetric about 0, expression (1) remains valid with $\Phi(x)$ replaced by $H(x) = \int_{-\infty}^{\infty} F(x + t)dF(t)$.

Finally, the mathematical argument guaranteeing the validity of the simultaneous confidence intervals in the parametric case cannot be applied to their rank transformation analog, even if one were interested in contrasts of the form $\sum c_i E\bar{R}_{i\cdot}$. The simultaneous confidence intervals would require

$$P\left(\left|(\bar{R}_{i\cdot} - \bar{R}_{k\cdot}) - (E\bar{R}_{i\cdot} - E\bar{R}_{k\cdot})\right| \le d_\alpha S\left(\frac{1}{n} + \frac{1}{m}\right)^{1/2} \text{ for all } i\right) \doteq 1 - \alpha, \quad (2)$$

regardless of the values of the θ's, where $S^2 = \sum\sum(R_{ij} - \bar{R}_{i\cdot})^2/(N - k)$ with $N = (k - 1)n + m$, and d_α is the multivariate t value for the two-sided Dunnett procedure with significance level α. Unfortunately, (2) is guaranteed to be asymptotically correct only when

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all the θ's are equal and the rank transformation analog of Shaffer's procedure can be used only to make the following more limited type of inference. If we declare a contrast different from 0 whenever the associated confidence interval does not include 0, then the probability of making any false declarations is controlled at α only when $\theta_1 = \theta_2 = \cdots = \theta_k$, an inference of questionable value.

The preceding example illustrates some of the difficulties that can arise when using rank transformations to develop new procedures. The example is not intended to show that all rank transformation procedures are bad, as this is certainly not the case. Rather, it illustrates that in some problems important properties of the original observations may not carry over to the ranks in the right way, and this can result in rank transformation analogs that solve uninteresting problems, are incorrect, or both. Thus, one must be careful when using rank transformations to develop new statistical procedures.

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Rejoinder

W.J. CONOVER AND RONALD L. IMAN

First, we would like to thank Gottfried E. Noether and Michael A. Fligner for taking the time to provide their valued perspectives to this subject of the rank transformation. Also, each discussant has added an original contribution to the field of rank transformation procedures that we hope will not be overlooked because of their locations as appendages to our article.

Noether presents a convincing argument in favor of a more careful presentation of nonparametric procedures, explaining the differences in the hypotheses from those of the usual parametric tests, showing how confidence intervals can be obtained, and essentially deriving the sampling distribution of the estimators and test statistics. To an expert in nonparametric statistics this seems like a simple enough course to follow. We wonder what the novice may think, however. These concepts, simple as they may seem to some, may be confusing to others. If a practitioner does not follow the logic as presented by Noether, he or she may not understand the pure mechanics of the procedure either. We may wish to recall John Tukey's argument for practical power as a basis for comparing tests, in which practical power is defined as the product of the mathematical power by the probability that the procedure will be used. The rank transformation provides a translation of the nonparametric concepts into the vocabulary of the practitioners, in the hope of increasing the probability that the nonparametric procedures will be used. We must agree that, as always, something is lost in the translation. Perhaps readers will be sufficiently interested in the translation, however, to want to follow what appears to be Noether's advice: read the original, untranslated version.

Our motivation for writing the article was to increase the visibility and usability of nonparametric techniques. That is, as Noether points out, many books now contain a separate chapter on nonparametric statistics; however, it is our observation that in many cases the addition of a chapter on nonparametric statistics is cosmetic in nature, added at the insistence of an editor or reviewer. The mere fact that a chapter is added does not mean it will be covered. Indeed, the additional chapter frequently appears at the end of the book and is commonly not covered for a variety of reasons. Even worse in our opinion is that segregating the chapter in this manner causes the student to lose the perception that nonparametric techniques provide alternatives to the parametric techniques when the assumptions are not satisfied. The rank transformation approach makes this transition easier by applying the parametric techniques to the ranks. This approach may hide the simplicity and greater applicability of the nonparametric approach, as Noether and Fligner point out. However, we feel that this is more than offset by enabling the student, as a matter of course, to check the assumptions of the parametric test rather

than apply them blindly under the misguided notion that the parametric procedures are robust and that one need not worry too much about the assumptions. In this vein, the rank transformation approach increases the likelihood that the nonparametric techniques will be covered and eases the burden on the instructor without a background in nonparametric statistics. At the same time, the instructor with an adequate background in nonparametric statistics can elaborate on the usefulness and general applicability of nonparametric techniques. In summary, we feel it is better to prepare the student to analyze a set of data and understand the assumptions and possible methods of analysis by presenting the parametric and nonparametric techniques side by side than it is to hope that students and instructors can bridge this gap on their own.

The discussion by Fligner emphasizes the point that the hypotheses being tested in the rank transformation procedure are not necessarily the same as those tested using the original data. This same point is valid for analyses on all types of transformed data, and it is often not even mentioned in analyses, say on the log transformed data. However, in a careful presentation the reader can become lost in discussions of precisely what is or is not being tested. Even the hypotheses for parametric tests may not be as clean as they first appear, because there are unproved assumptions hidden in the background, which may contribute to the acceptance or rejection of the hypothesis being tested if not really valid. Discussion of the hypotheses being tested can be much more involved than a casual observer may realize. We do not dispute, however, the contention of both discussants that the rank transformation procedure does test a different hypothesis than that tested by the corresponding parametric procedure. We should have made that point clear at the beginning of the article.